

Forest Fire Prevention via Simulation and Detection using Computer Vision



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INTRODUCTION & THESIS STATEMENT

Nowadays, forest fires are one of the most serious natural hazards that affect in numerous aspect the environment and people's lives. Even though many measures have already been taken and the technology is progressing, forest fires remain a menace in the world.

Forest fire early detection through Artificial Intelligence has proven to be the most effective measure for fire prevention. Therefore, *It is possible to develop a CNN based system for image classification for the detection of fire and smoke.*

OBJECTIVES

1. Develop a simulated forest environment with various environmental conditions having different fire and smoke patterns using UNITY
2. Create a dataset for fire and smoke multi-class classification from the simulated environment
3. Create second real-world dataset from video frames to train the models
4. Implement CNN models using Computer Vision to train it on the train dataset
5. Evaluate and compare the models' performance
6. Document and demonstrate the results

MOTIVATION FOR THIS PROJECT

The motivation for this project was clearly from an environmental aspect. Additionally, early fire detection reduces human, environmental, and economic losses, which I also aim to minimize. That is why my goal with this thesis is to develop and evaluate efficient CNN-based models for accurate multi-class fire and smoke classification, focusing on architectures that are suitable for deployment in practical monitoring devices such as drones.

DATASET DESCRIPTION

For this project two authentic datasets were created, one simulated and one from real-world videos.

UNITY SIMULATED DATASET

- Created inside a custom-built in Unity environment designed to simulate European forest areas where fires might occur.
- Used the same environment to create a vast variety of simulations in different case scenarios through a dynamic way of combining terrain assets, tree meshes, terrain contour lines, and other environmental features.

REAL-WORLD DATASET

- Limitations of the synthetic dataset led to a necessity of a real-world dataset.
- For this purpose, a new dataset was created by taking frames of fire and smoke video footage, from Envato Elements, a licensed stock media platform.
- From this dataset a second version was created to resolve the issues of black smoke lack in early fires. This dataset was the one used in the training process, containing 5 classes and approximately 6000 samples.

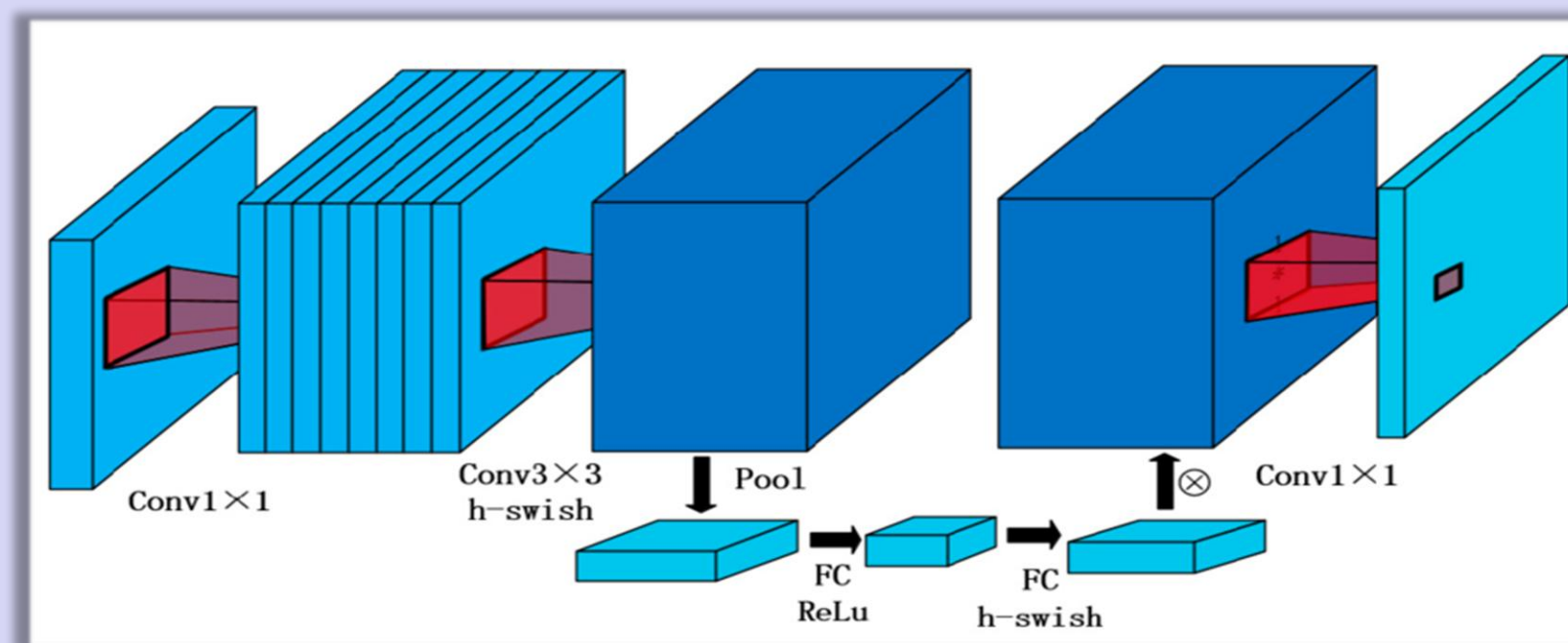
LITERATURE REVIEW

Machine Learning:

- A branch of Artificial Intelligence that is applied in areas such as Computer Vision
- Allows computers to learn from the data they collect, thus recognizing patterns while not being explicitly programmed for each task, to predict and adapt to new data, make their own decisions, improving their performance over time

Models:

- ConvNextTiny: a Machine Learning model with a modern convolutional neural network architecture that offers high performance in image classification with reduced computational cost
- MobileNet-v3-Large: a Machine Learning model with a Convolutional Neural Network (CNN) architecture suitable for mobile or edge system applications, i.e. with limited resources for high-performance image classification quickly and accurately



- ResNet-50: a popular CNN for computer vision that is trained without the problem of signal vanishing in deep networks, it is 50 layers deep and is widely used for image classification
- EfficientNet-B0: Google's smallest and most successful EfficientNets model for computer vision that delivers accuracy and performance with limited computational resources
- BEiT: a self-supervised vision representation model (Bidirectional ENCODER REPRESENTATION FROM IMAGE TRANSFORMERS), which using the "masking" technique, performs Deep Learning visual tokens from images

Unity:

- One of the most well-known development platforms through which complete realistic environments and simulations are created in the digital world

Datasets:

- Large organized data sets (e.g. images, videos)
- Usually labeled into categories
- Used to train and evaluate the algorithm with knowledge and skills that it should have
- Have the most important role in its generalization, reliability and productivity

CONCLUSION & FUTURE WORK

In conclusion the task of producing a system that can classify fire and smoke based on custom datasets with images can prove challenging. However, effective solutions can be achieved using compact models suitable for real-time deployment.

Main Steps:

- Creation of datasets (synthetic via Unity and real-world from aerial videos)
- Initial experiments and dataset versioning
- Data augmentation and preprocessing
- Fine-tuning and evaluation of multiple CNN models

Improvements:

1. Improvements in datasets and in the creation of synthetic dataset and its labeling
2. More experiments and more fine-tuning of best models

RESULTS & DISCUSSION

Each model's evaluation consists exclusively of an objective quantitative evaluation. Specifically:

- Multiple CNN and transformer-based models evaluated

- Best unseen-set accuracy: MobileNet - 78.32%
- Best fine-tuned model: ConvNeXt (3 frozen layers)
 - Accuracy: 83.42%
 - Avg F1: 85.28%
 - Weighted Precision: 90.09%

- K-fold cross-validation confirms robustness of lightweight models

Discussion

- Fire and smoke classification is challenging due to class similarity
- Large models and transformers tend to overfit
- Lightweight architectures (MobileNet, ConvNeXt) perform best
- Suitable for real-time drone deployment (e.g., NVIDIA Jetson Orin)

SELECTED BIBLIOGRAPHIC REFERENCES

1. P. Barmpoutis, P. Papaioannou, K. Dimitropoulos, and N. Grammalidis, 'A review on early forest fire detection systems using optical remote sensing', *Sensors*, vol. 20, no. 22, p. 6442, 2020, Accessed: Dec. 05, 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/20/22/6442>
2. A. M. Al-Ansi and A. Al-Ansi, 'An overview of artificial intelligence (AI) in 6G: Types, advantages, challenges and recent applications', *Bul. Ilm. Sarj. Tek. Elektro*, vol. 5, no. 1, pp. 67-75, 2023.
3. G. Allen, 'Understanding AI technology', 2020, Accessed: Dec. 05, 2025. [Online]. Available: <https://apps.dtic.mil/sti/html/tr/AD1099286/>
4. B. Mahesh, 'Machine Learning Algorithms - A Review', *Int. J. Sci. Res. IJSR*, vol. 9, no. 1, pp. 381-386, Jan. 2020, doi: 10.21275/ART20203995.
5. I. H. Sarker, 'Machine Learning: Algorithms, Real-World Applications and Research Directions', *SN Comput. Sci.*, vol. 2, no. 3, p. 160, May 2021, doi: 10.1007/s42979-021-00592-x.
6. K. Sharifani and M. Amini, 'Machine learning and deep learning: A review of methods and applications', *World Inf. Technol. Eng. J.*, vol. 10, no. 07, pp. 3897-3904, 2023, Accessed: Dec. 05, 2025. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4458723